

V810i S2EX Phase Shift Profilometry 2 (PSP)

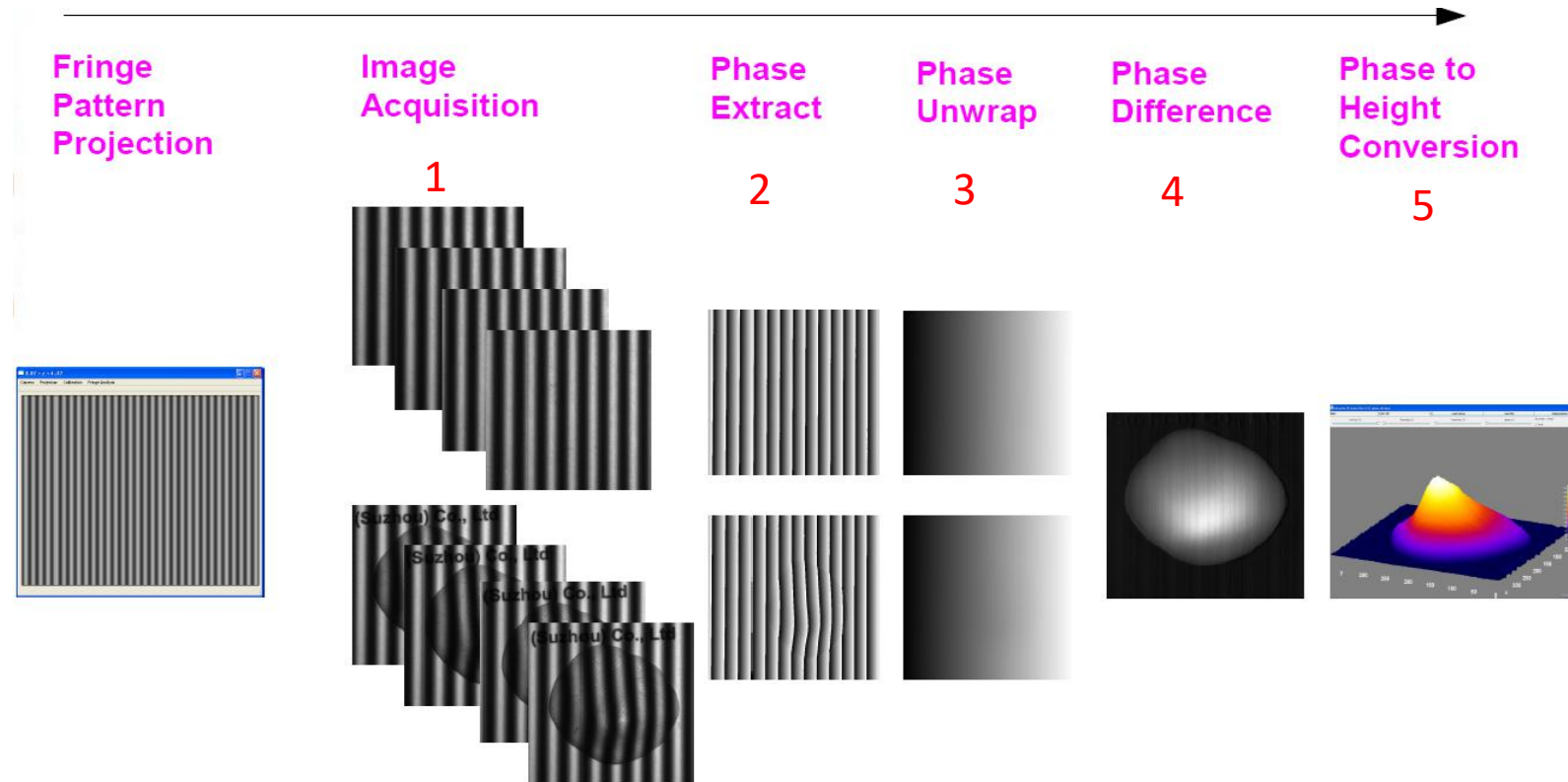


Introduction

The objective of Phase Shift Profilometry (PSP) is to solve the 100% PTH and Press fit Printed Circuit Assembly (PCA) which most of current AXI system could not inspect with high repeat-ability (except Agilent's legacy 5DX system).

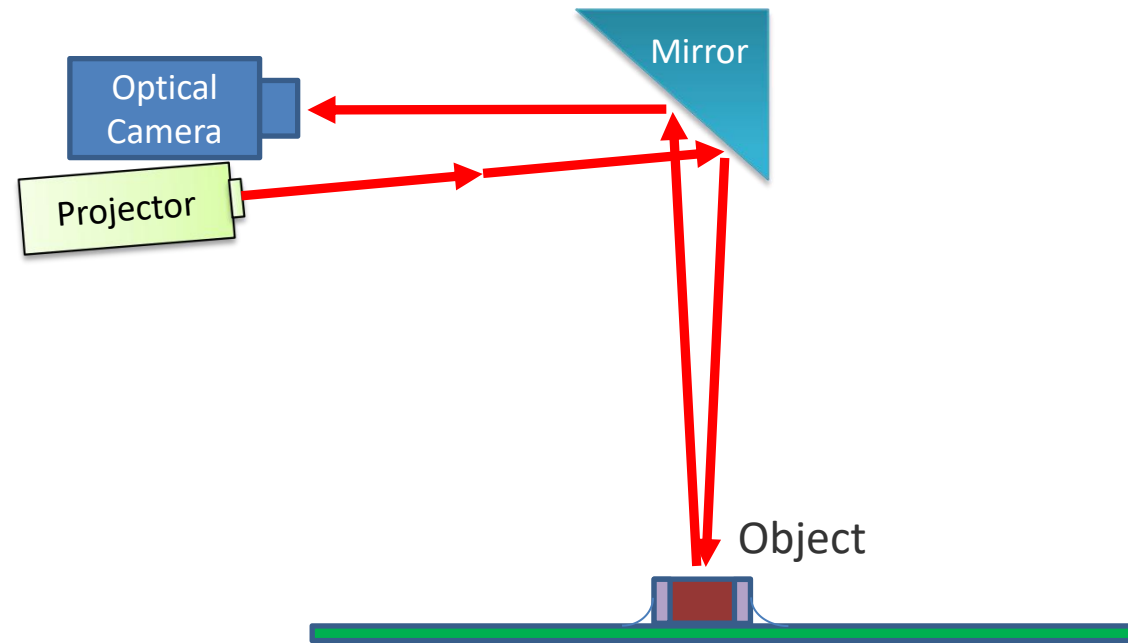
Moreover, some of difficult-to-focus components like LGA, micro SMT connectors are also use PSP for better focusing.

Phase Shift Profilometry (PSP) Sequence used to plot board surface



- A known height calibration plate is used as a reference for comparison with the object taken. The actual 3D image will then be acquired by using the different in phase shift of the object and calibration plate.

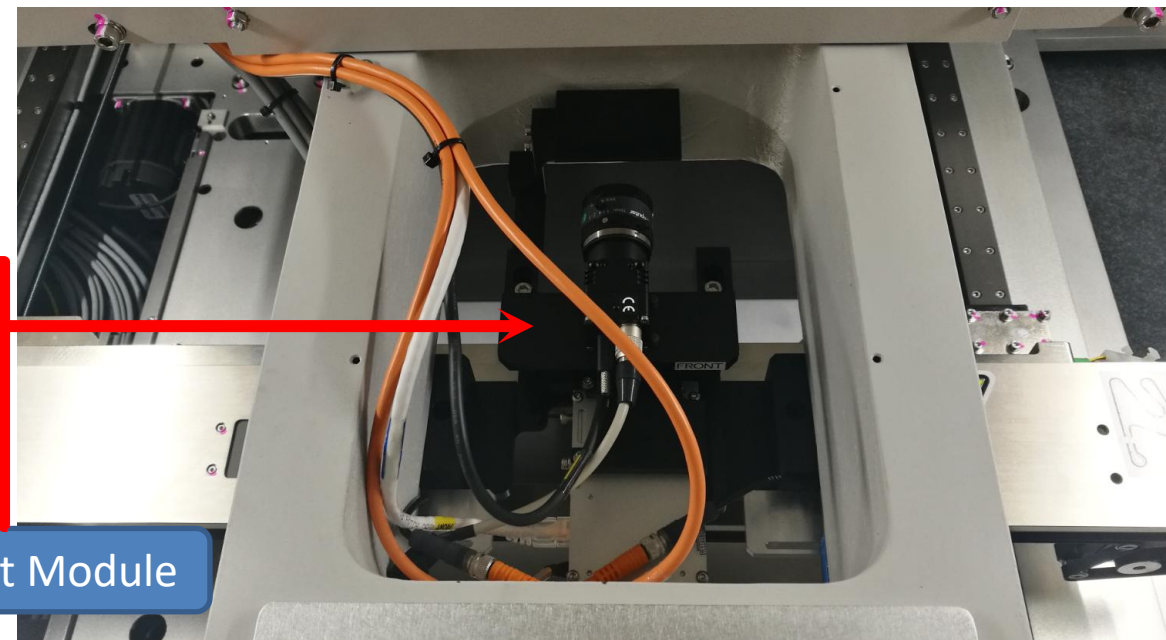
Schematic diagram of the proposed 3-D surface Profilometer



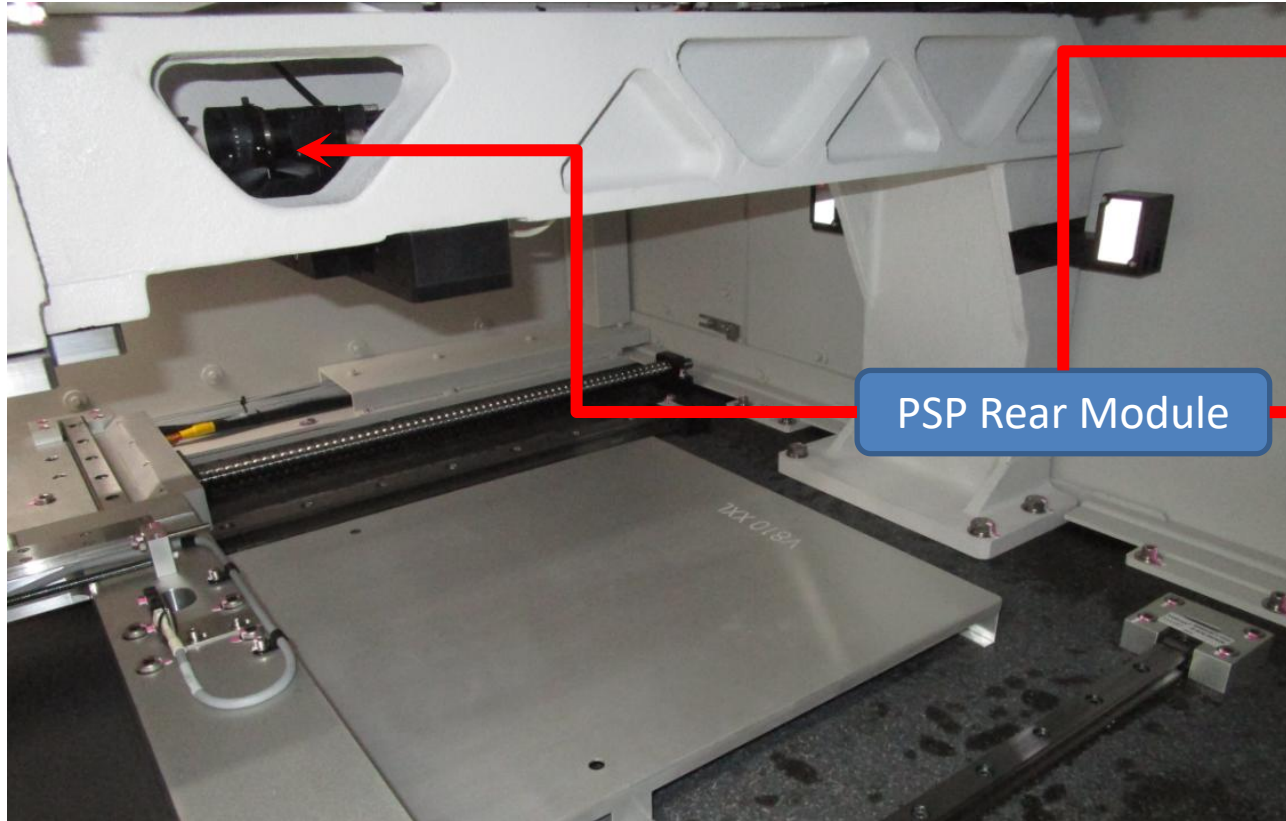
- A basic hardware that consists of projector and camera setup is needed for PSP.



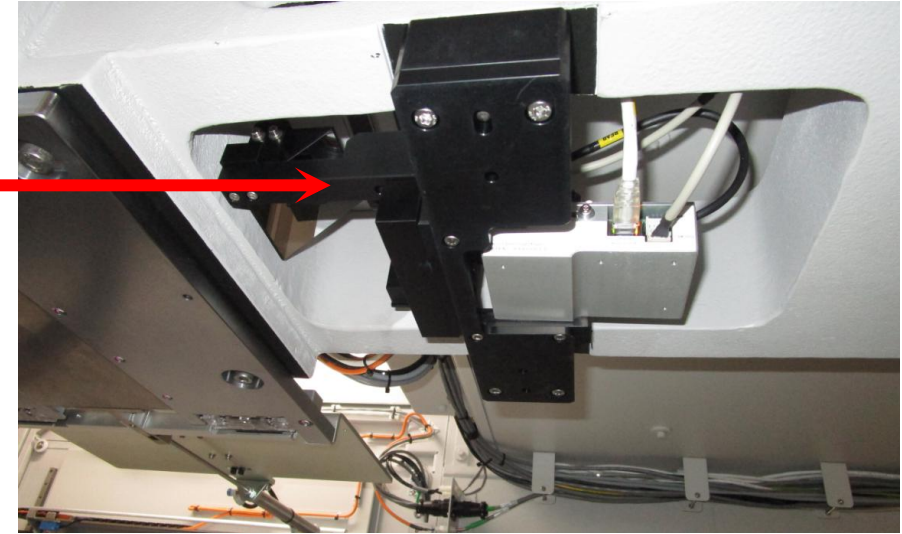
PSP Front Module



Front Door view

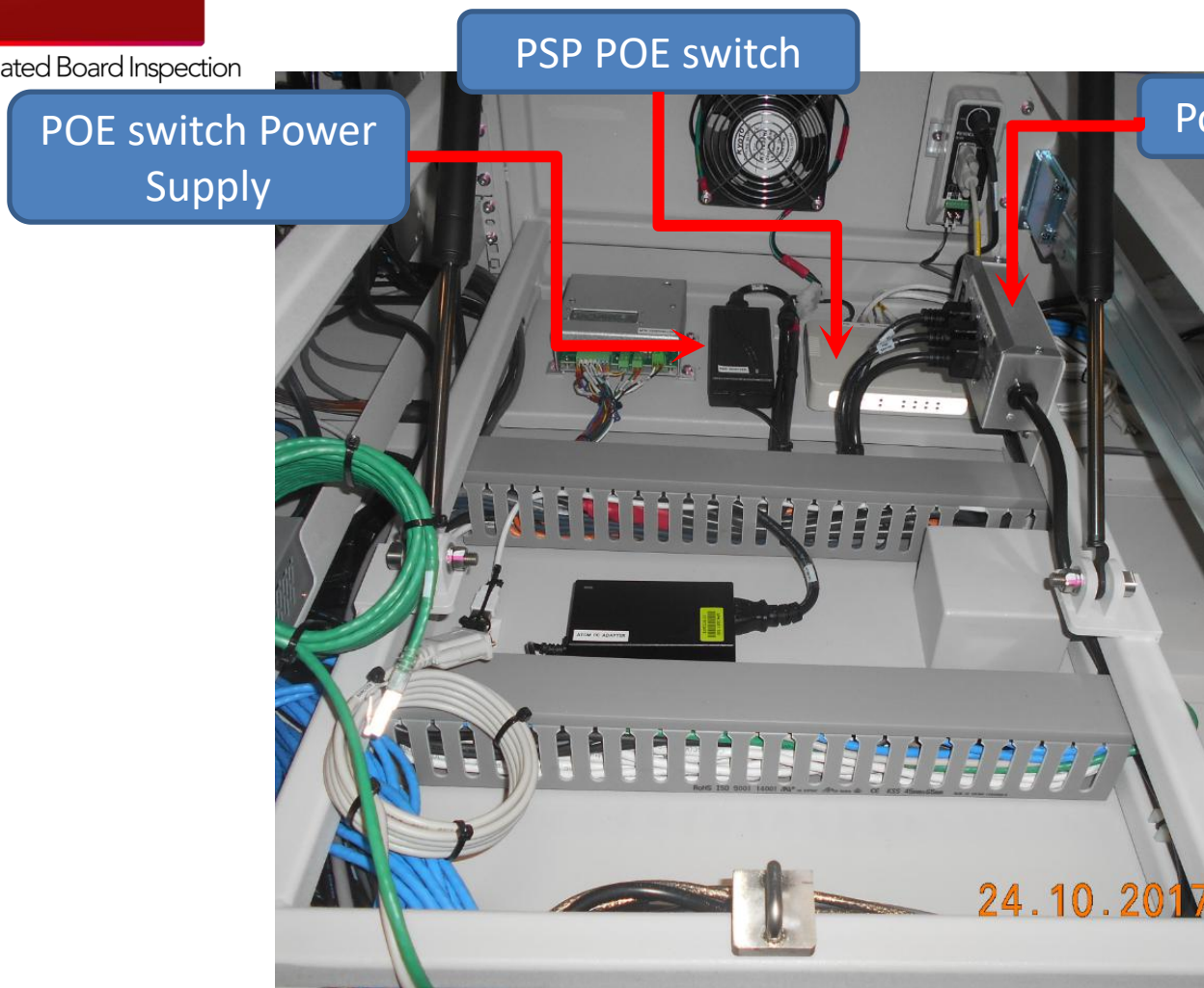


Right Outer Barrier view



ABI V810i GUI - PSP Hardware Overview

Automated Board Inspection

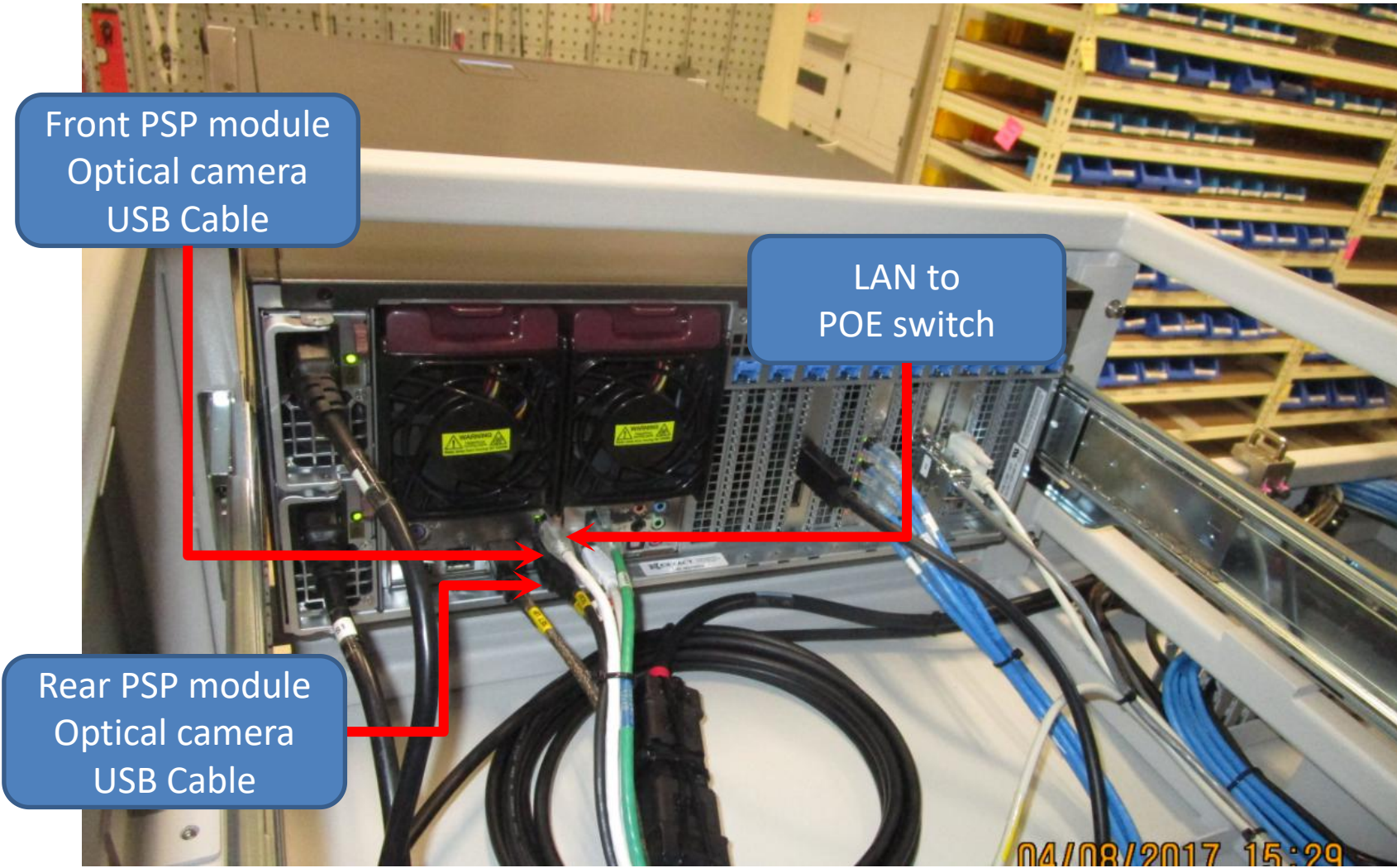


Server Rack view

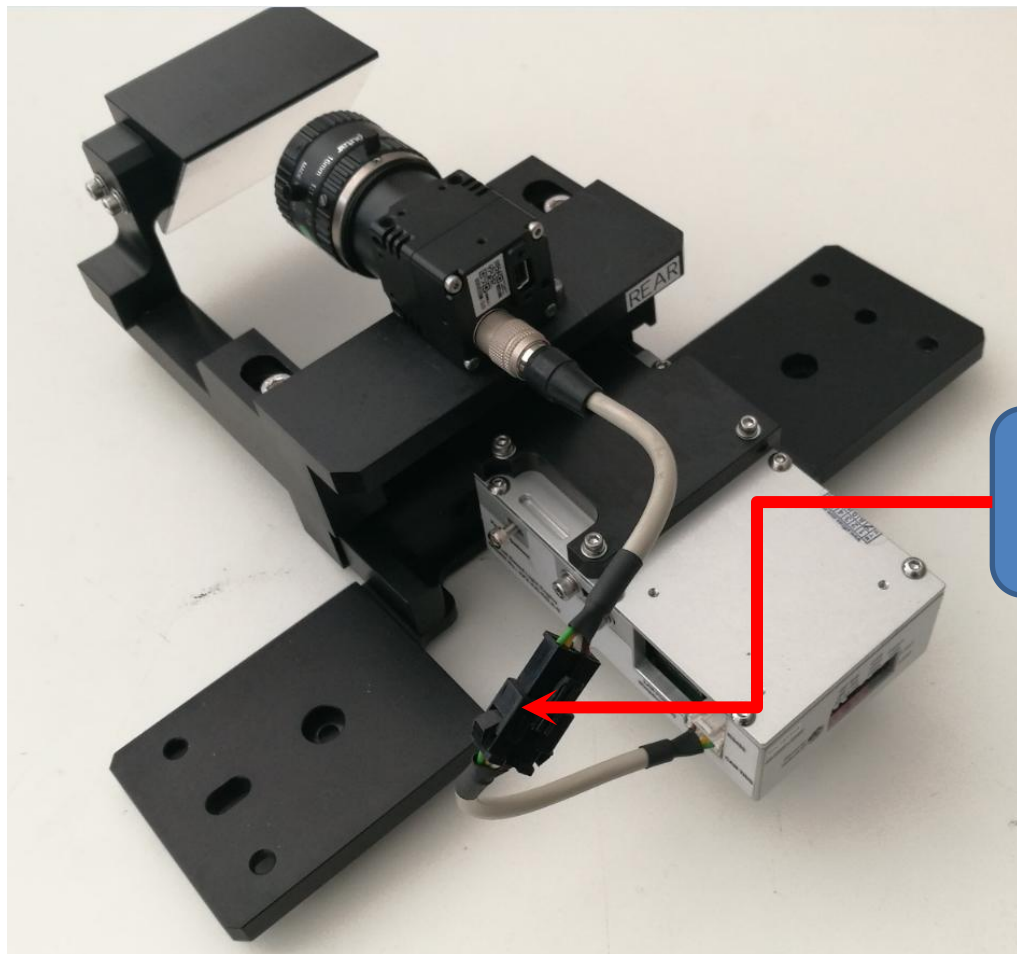
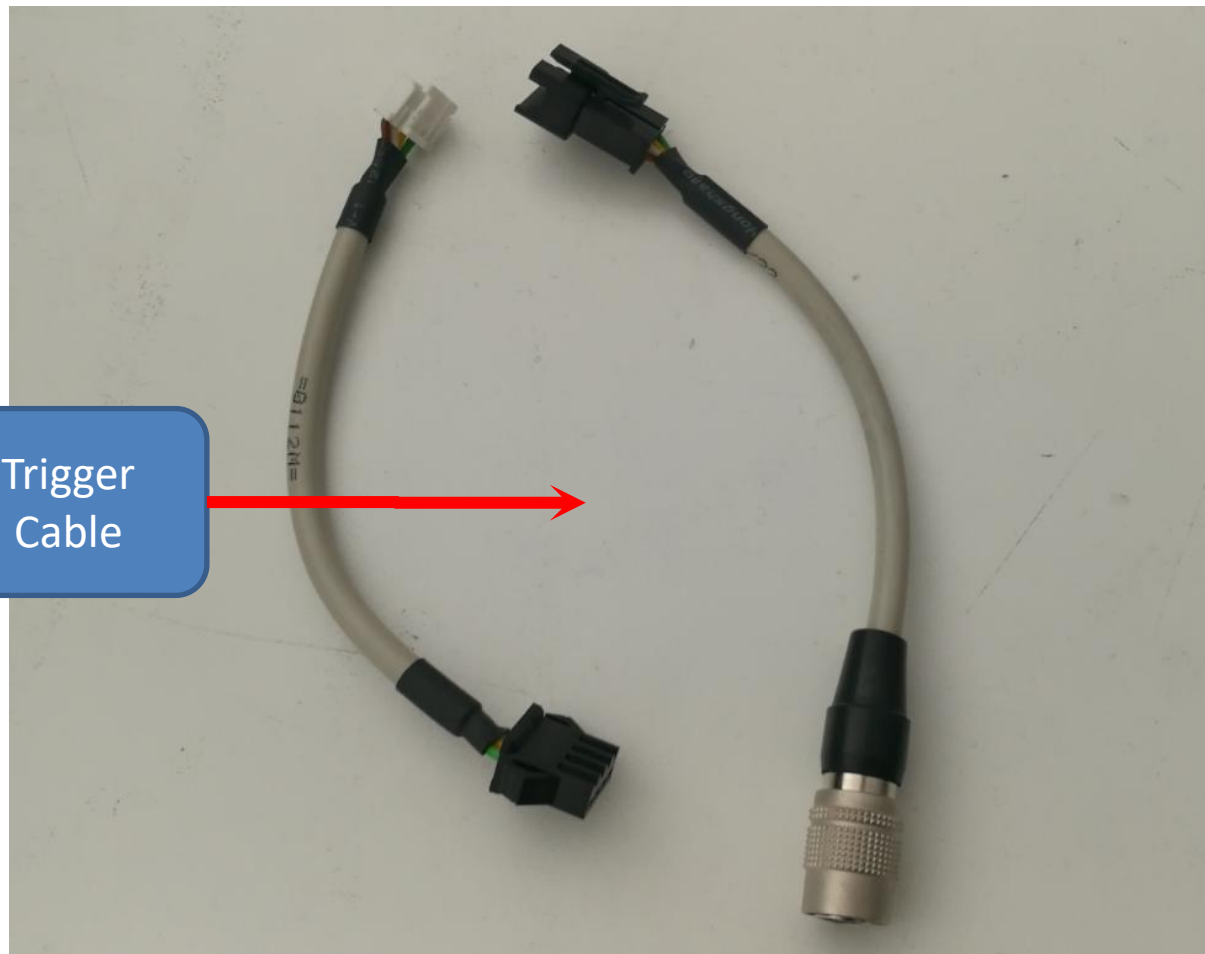


V810i GUI - PSP Hardware Overview

Automated Board Inspection



Server Rack view

Trigger
Cable

Power Adapter

Input: 240vAC, 1.5A 50/60Hz
Output: 48vDC, 1.5A

POE Switch

240VAC UK Power cord

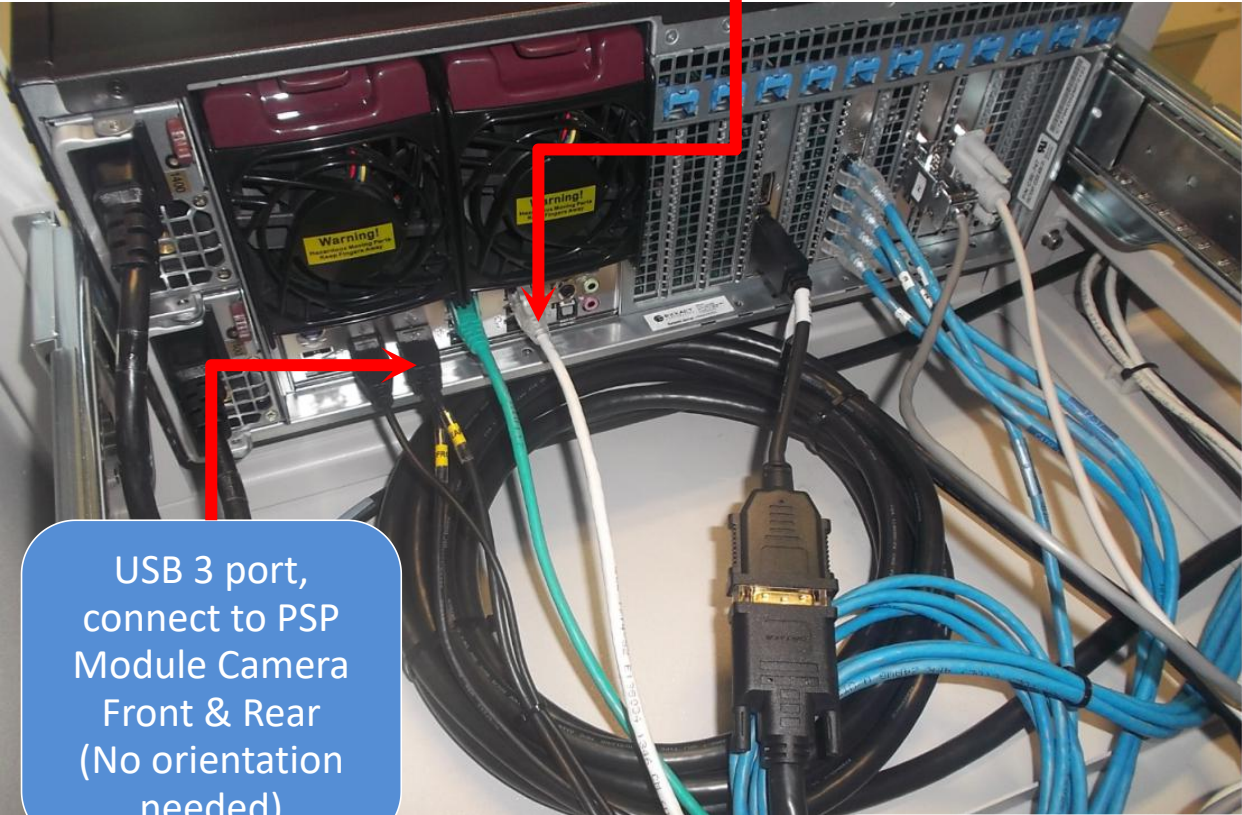




V810i PSP 2 Hardware Overview – POE Switch

Automated Board Inspection

Connect to
POE Switch

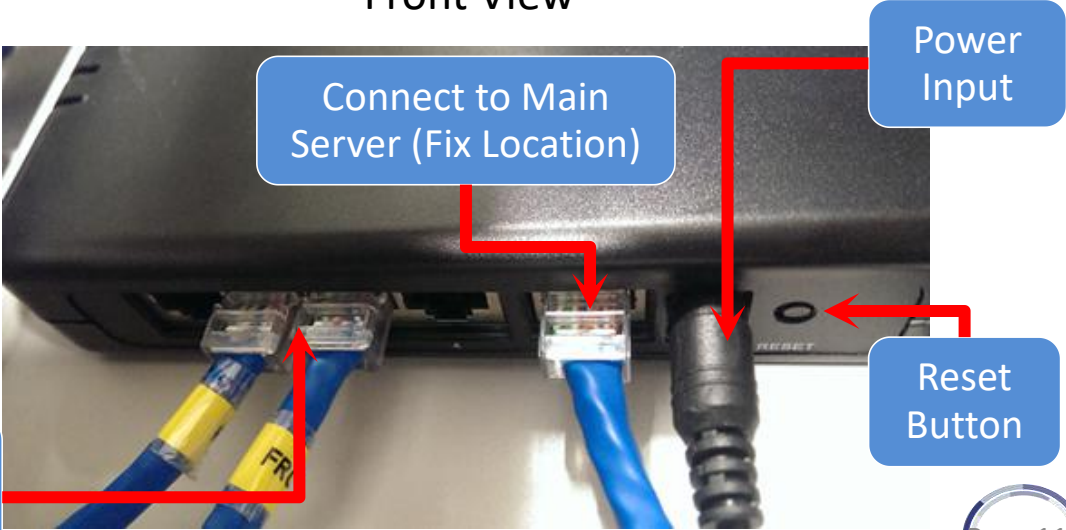


USB 3 port,
connect to PSP
Module Camera
Front & Rear
(No orientation
needed)

Connect to PSP Module
Projector Front & Rear
(No orientation needed)



Front View



Connect to Main
Server (Fix Location)

Power
Input

Reset
Button

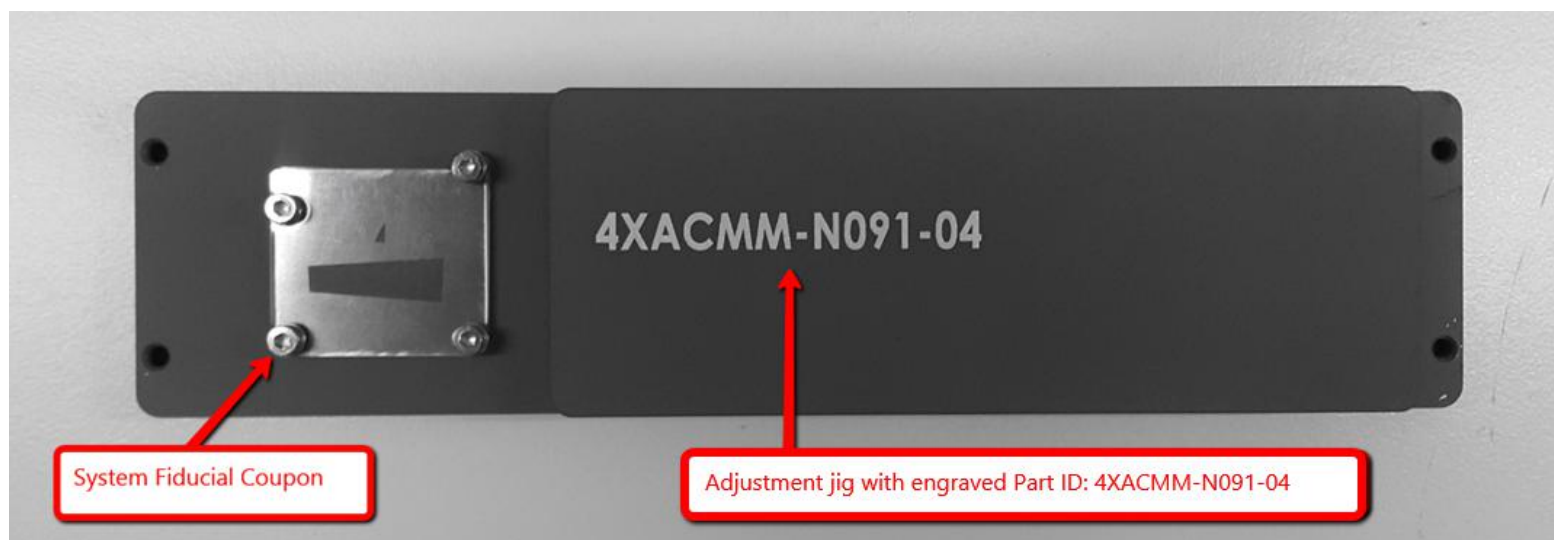
Rear View



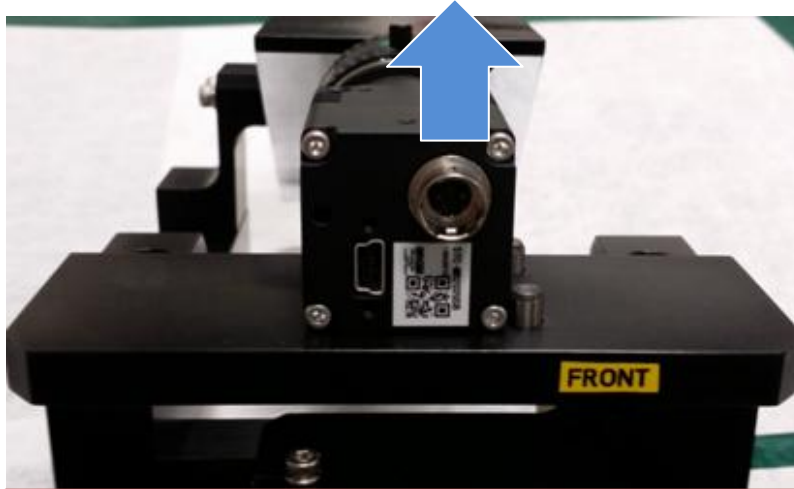
V810i PSP 2 Hardware Overview – Manual Load Jig

Automated Board Inspection

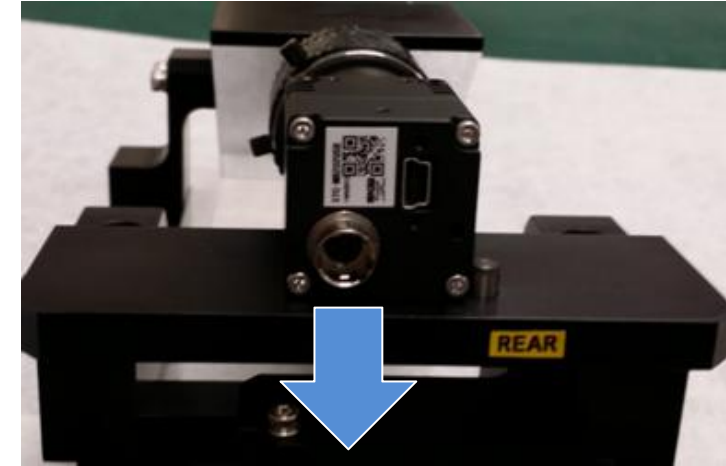




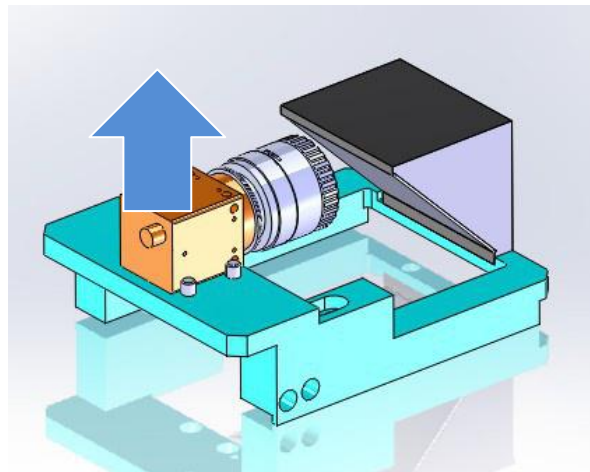
Front VS Rear PSP Module - Camera Orientation



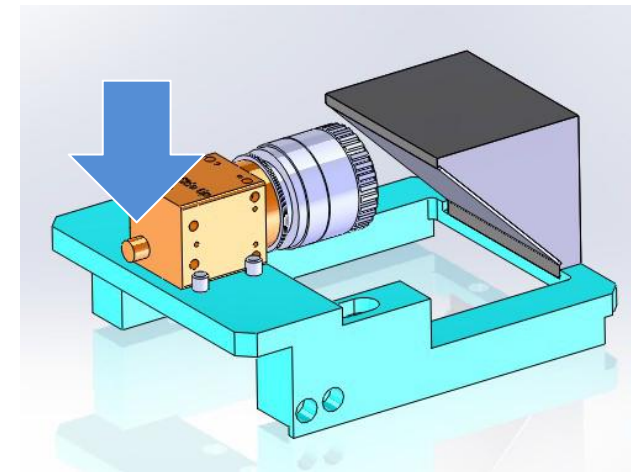
Connector face TOP



Connector face Bottom



Front Camera Orientation



Rear Camera Orientation



Front VS Rear PSP Module - Projector IP Address Setting

Automated Board Inspection



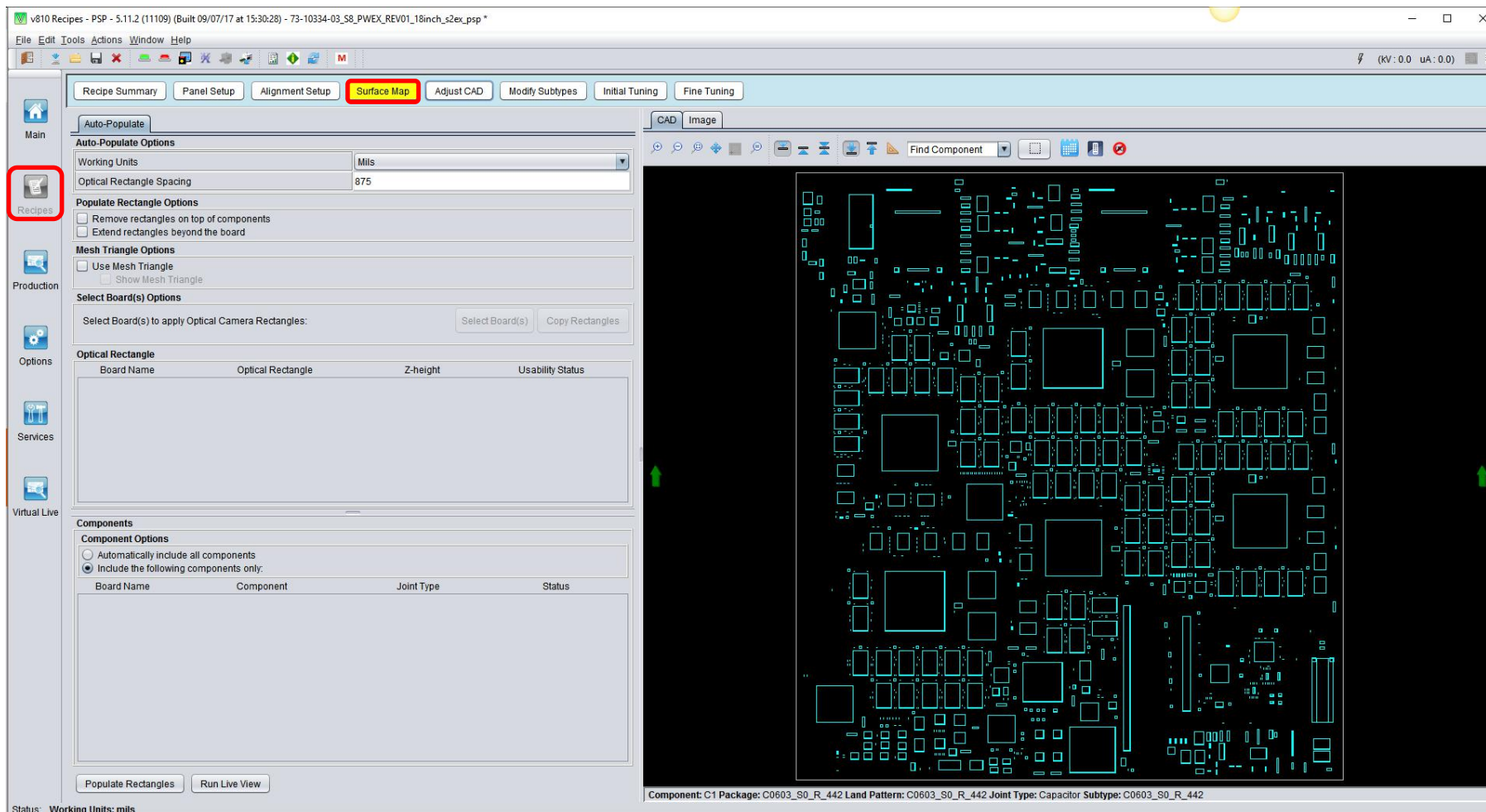
Front Projector IP setting – 123.123.123.100

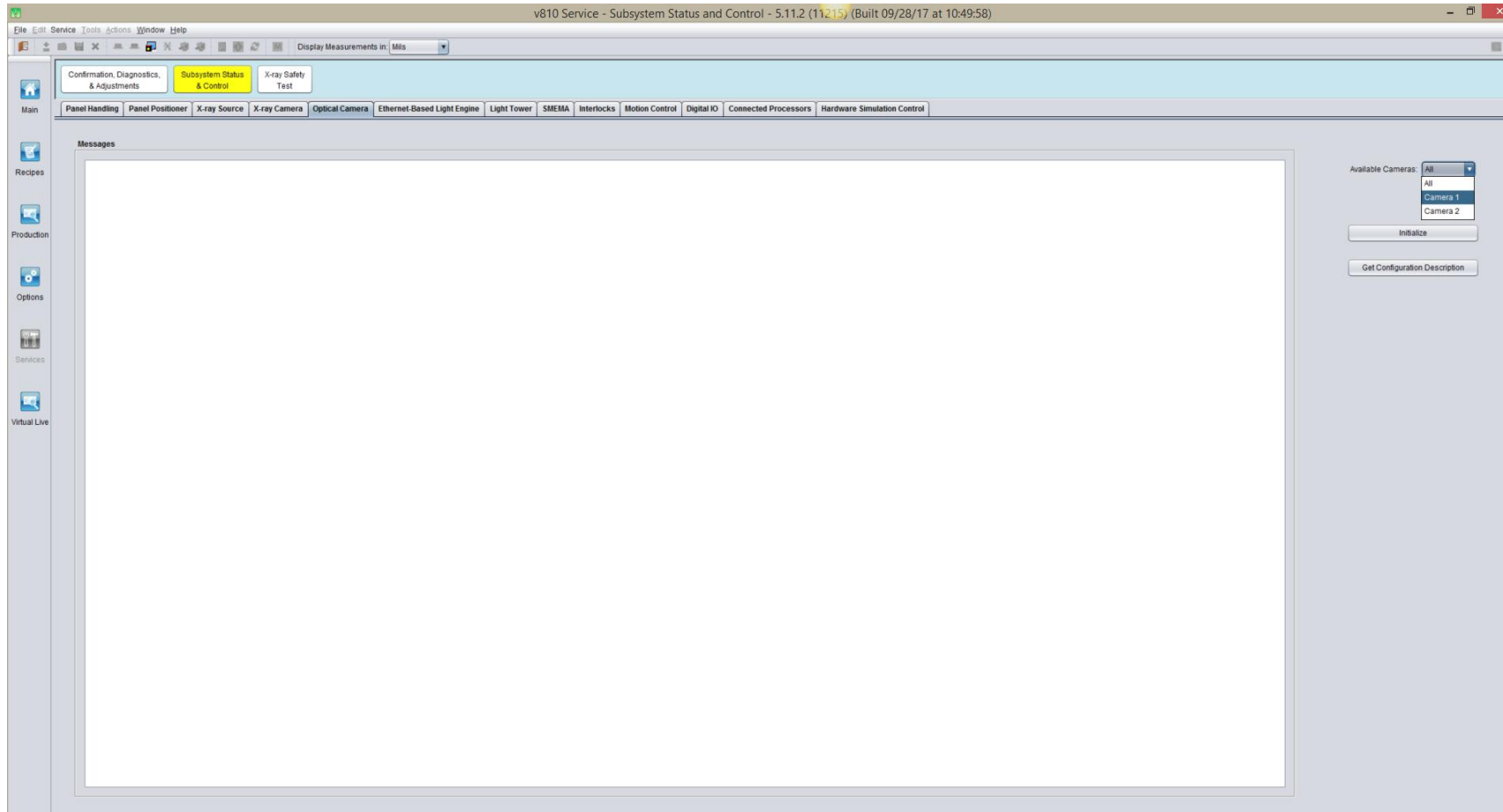


Rear Projector IP setting -123.123.123.101

ABI V810i GUI - PSP Setup Interface

Automated Board Inspection



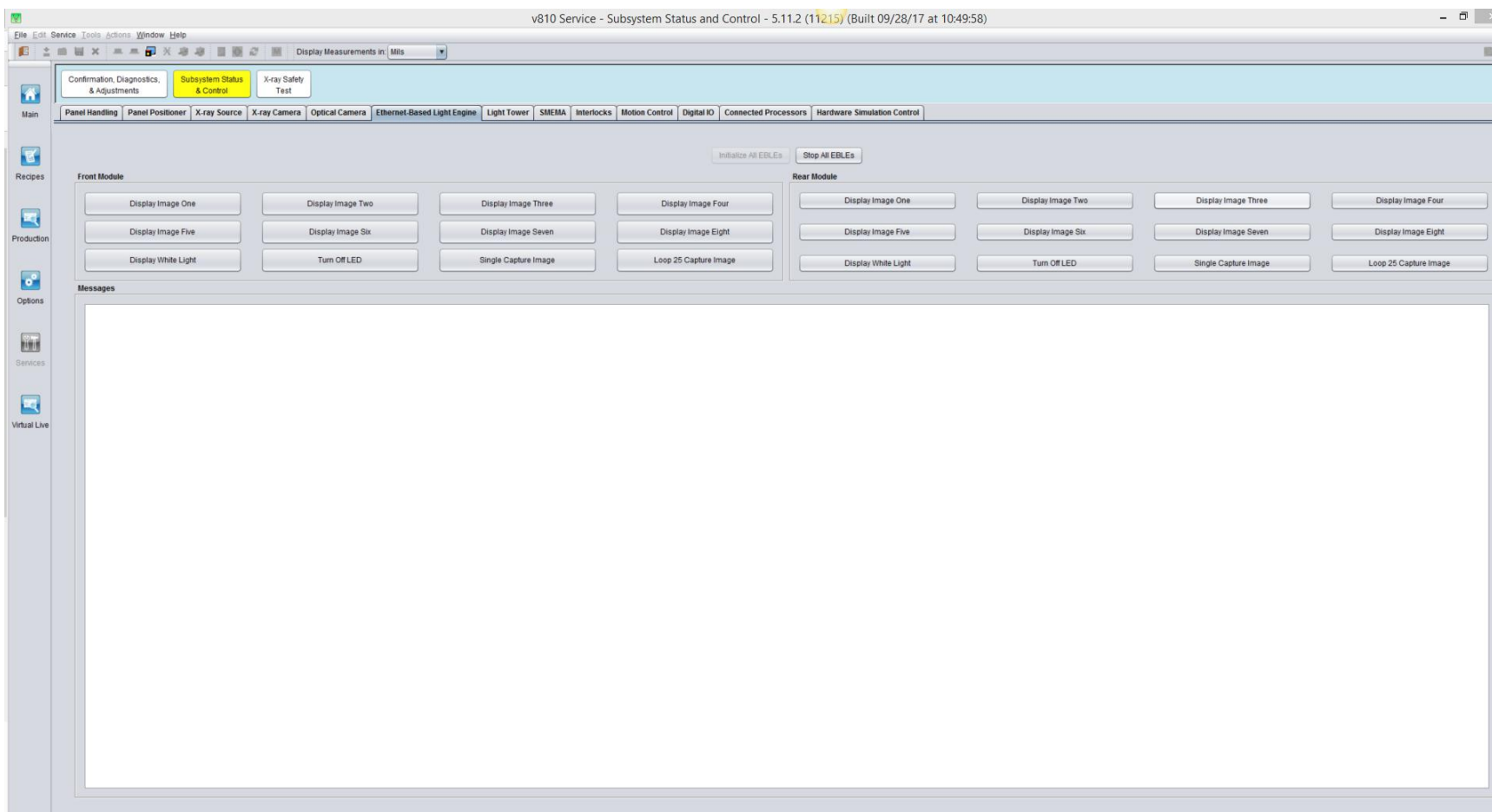


- Communication status verification for PSP optical camera with main server
- Unable initialize if hard ware failure.
(# Root Cause; USB cable not seat firmly or optical camera power lost)

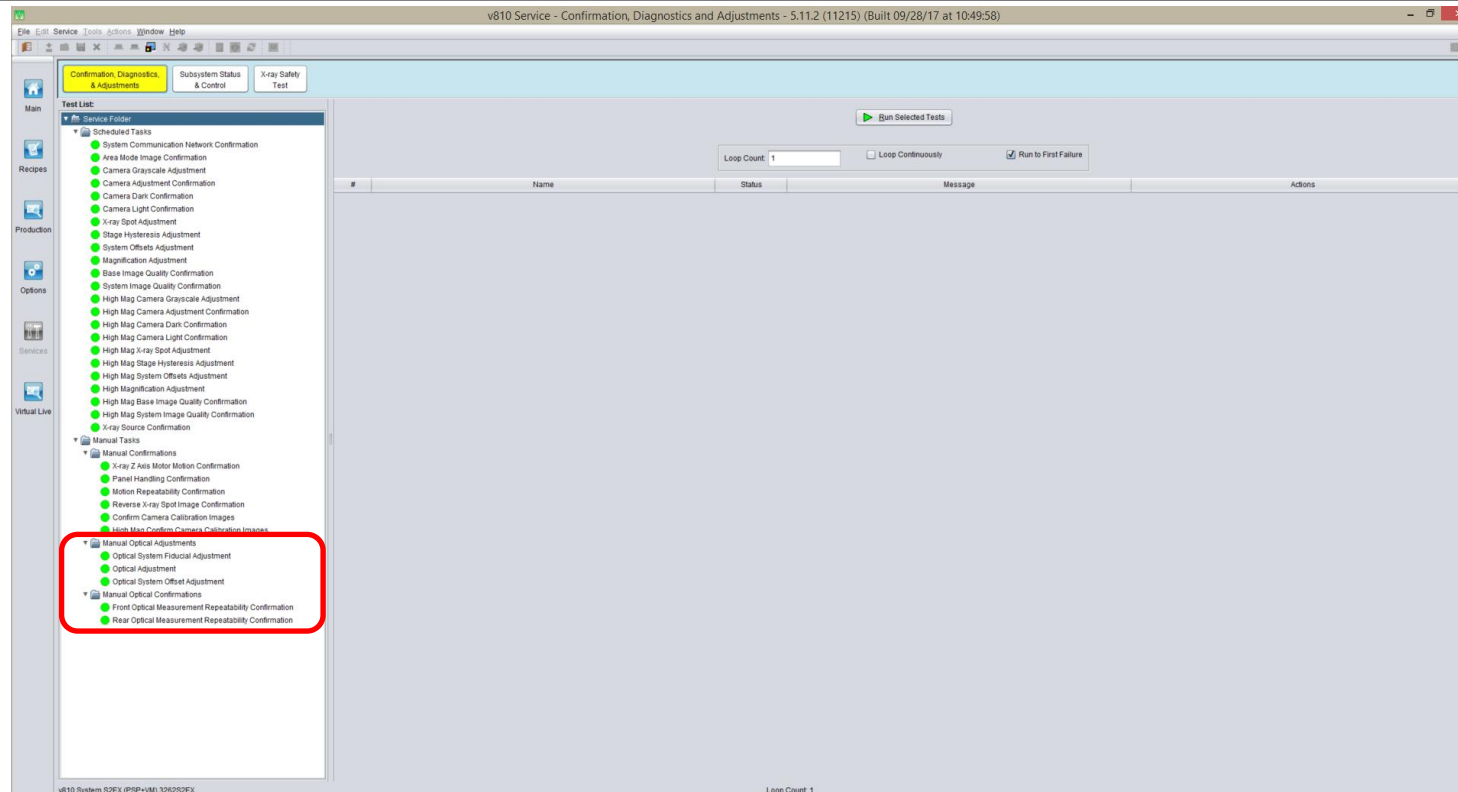


Automated Board Inspection

V810i PSP 2 User Interface – EBLE Projector tab



- PSP 2 projector main tab
- Manually trigger project with selected fringe pattern for focusing verification
- Unable initialize if power lost at projector or trigger cable connection issue



There will be 5 task to be performed during at CD&A

- Optical System Fiducial Adjustment
- Optical Calibration Adjustment
- Optical System Offset Adjustment
- Front Optical Measurement Repeat-ability Confirmation (SW Ver 5.8 and above)
- Rear Optical Measurement Repeat-ability Confirmation (SW Ver 5.8 and above)

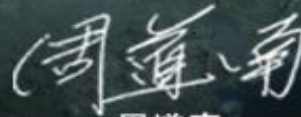
Remark: Above tasks will not automated running while machine start-up and need manual trigger



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